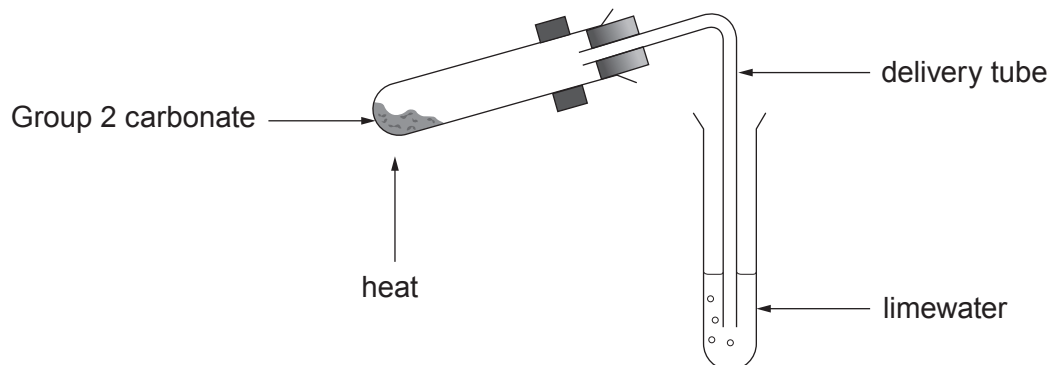


10. (a) A student investigated the thermal stability of Group 2 carbonates. She used the following apparatus and method.



1. Start a stopwatch at the moment you begin to heat the carbonate and continue to heat for 4 minutes or until the limewater turns cloudy.
2. After the limewater turns cloudy lift the delivery tube out of the limewater then remove the flame from under the boiling tube.
3. Repeat the heating procedure for each carbonate in turn.
4. Use a spatula-measure of the appropriate carbonate, fresh limewater and heat with the hottest Bunsen burner flame each time.

Her results are shown below.

Carbonate	Time taken for limewater to turn cloudy/s
MgCO ₃	20
CaCO ₃	40
SrCO ₃	230
BaCO ₃	does not turn cloudy



- (i) Suggest an improvement to the method to ensure that the experiment is a fair test. [1]

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- (ii) Suggest why the delivery tube should be lifted out of the limewater before the flame is removed from under the boiling tube. [1]

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- (iii) State what conclusion she can draw about the thermal stabilities of the Group 2 carbonates from these results. [1]

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- (iv) The student was told that the temperature at which barium carbonate decomposes is 1360°C . The maximum temperature of a typical Bunsen burner flame is around 800°C .

State whether the limewater would have turned cloudy if she had used two Bunsen burners to heat the barium carbonate. Give a reason for your answer. [1]

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- (b) Barium nitrate also decomposes on heating.



- (i) In an experiment 0.960 g of barium nitrate was heated strongly for 2 minutes.

Calculate the maximum volume, in cm^3 , of gas that could be produced at a temperature of 25°C and a pressure of 1 atm.

[3]

Volume = cm^3

- (ii) The volume of a gas is directly proportional to its temperature at constant pressure.

A student said that if the gas formed in this experiment were collected at a temperature of 50°C and at 1 atm pressure, the volume formed would be double that calculated in part (i).

Do you agree? Justify your answer.

[1]

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- (c) State the conditions necessary for **each** of barium oxide and barium metal to conduct electricity. Explain this property in terms of structure and bonding in each case. [3]

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- (d) The atomic radius of a barium atom is 0.217 nm.

From the list below, choose the value for the **ionic** radius of a barium ion. Give a reason for your choice. [2]

0.135 nm

0.210 nm

0.217 nm

0.265 nm

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- (e) A sample of barium contains two isotopes. The first isotope has a relative isotopic mass of 134.9 and the second a relative isotopic mass of 137.9. The relative atomic mass of the sample of barium is 137.3.

Calculate the percentage abundance of the first isotope. [2]

Abundance = %

